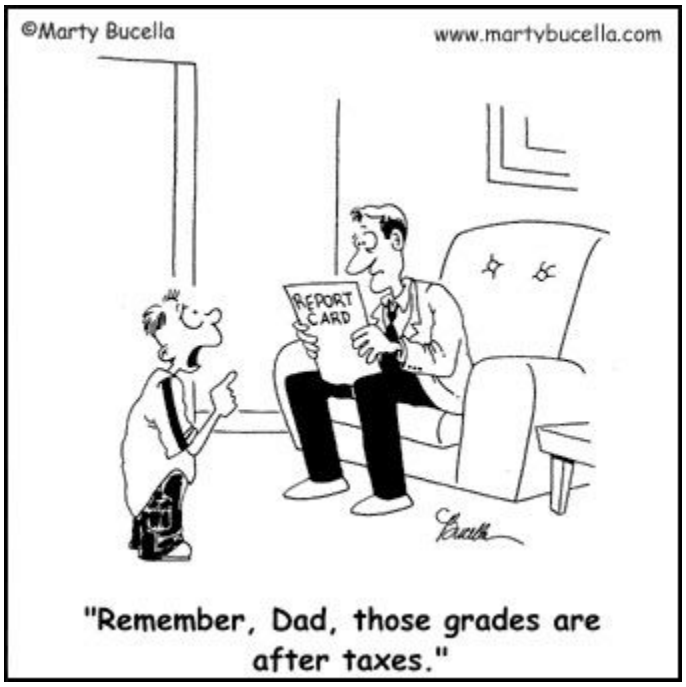
The Wharton School
Accounting 6130 - Lane
Final Exam
SOLUTION
December 5, 2025

PROBLEM	MAXIMUM	SCORE
I. Long-term Liabilities	27	
II. Income Taxes	21	
III. Stockholders' Equity	26	
IV. Cost Concepts	30	
V. Transfer Pricing	16	
TOTAL	120	



"Remember, Dad, those grades are after taxes."

III. Long-term Liabilities (27 points)

Refer to Estee Lauder Companies, Inc.'s 2024 financial statement excerpts to answer the following questions. **Note that the fiscal year 2024 is the year ended June 30, 2024.**

1. How much of Estee Lauder's long-term debt is expected to be paid in **fiscal year 2025**? (3 points)

   $504, current maturities of long-term debt

   See the Balance Sheet or Note 12

2. Which of the following are true about mortgage accounting and bond accounting? Circle all that apply. (3 points)

   a. The borrower and lender agree upon the borrowing rate for both debt instruments
   b. Both debt instruments are amortized to zero as of the maturity date of the debt
   c. Interest expense is the outstanding borrowing times the true borrowing rate (the effective rate)
   d. The cash payment does not change over the life of the debt
   e. The book value of the debt is equal to the present value of the debt on the balance sheet.

3. Consider the 2.600% Senior Notes due April 15, 2030 (i.e., the 2030 Senior Notes). Were these bonds issued at par, a discount, or a premium? Circle one. (2 points)

   a. Par
   b. Discount
   c. Premium

4. Consider again the 2.600% Senior Notes due April 15, 2030 and assume that the fair value of these 2030 Senior Notes was $700 million at June 30, 2024. Based on this assumption answer parts A and B below.

   A. How did the market rate of interest change, if at all, since the issuance of these notes? Circle one. (2 points)

   i. The interest rate increased
   ii. The interest rate decreased
   iii. The interest rate did not change

B. What gain or loss would Estee Lauder recognize on these notes if they retired (i.e., repurchased) the notes on June 30, 2024? Indicate if it is a gain or a loss and the amount. If no gain or loss, write NONE. (4 points)

Answer: ___**105**_____ Gain **Loss**

They would have to pay $700 million to retire debt with a book value of $595 (See Note 12). Thus they would recognize a **loss in the amount of the difference of $105**. You can see that only the loss would balance the journal entry to record the repurchase.

Dr. Sr. Notes	595	
Dr. Loss	**105**	
Cr. Cash		700

5. Consider the June 2024 issuance of the 5.000% Senior Notes due June 30, 2034 (the 2034 Senior Notes), and assume these notes were issued on June 30, 2024 (i.e., at the very end of the fiscal year). Record the journal entry Estee Lauder **made at issuance**. (5 points)

Amount of cash received is face value x price = $650 million x .99689 = $647,978 thousand (See 2nd table of Note 12)

Dr. Cash	**647,978**	
Cr. 5.0% Sr. Notes		**647,978 (OK to use 648 as the amount)**

OR

Dr. Cash	**647,978**	
Dr. Discount	**2,022**	
Cr. 5.0% Sr. Notes		**650,000**

6. Considering again the Senior Notes due 2034, what is the summary journal entry for the first year of six months interest payment? (6 points)

Outstanding borrowing for 1st six months is the amount received at issuance
Interest expense = outstanding borrowing x discount rate
= 647,978 x .0504/2 = 647,978 x .0252 = 16,329

Dr. Interest Expense	**16,329**	
Cr. Discount or Sr. Notes		**79**
Cr. Cash		**16,250**

Cash payment = face value x coupon rate = 650,000 x .05/2 = 16,250

7. Per Note 17, Estee Lauder has purchase obligations of $2,370 million as of June 30, 2024. How much of these obligations are on Estee Lauder's June 30, 2024 Balance Sheet? Circle one. (2 points)

   a. $2,370 million
   b. $1,463 million
   c. $98 million
   d. $0

   Purchase obligations are executory contracts and are off-balance sheet liabilities.

IV. Income Taxes (21 points)

Refer to Estee Lauder Companies, Inc.'s 2024 financial statement excerpts to answer the following questions. **Note that the fiscal year 2024 is the year ended June 30, 2024.** Assume that Estee Lauder's deferred tax assets and deferred tax liabilities are calculated using only **the federal statutory tax rate of 21%.**

1. How much did Estee Lauder owe to governments for income taxes in fiscal year 2024? (2 points)

   $628

   The current portion of taxes; See the first table of Note 9

2. What was the change in Deferred Taxes due only to operations? (3 points)

   $265 decrease or debit

   See the Deferred portion of taxes in the first table of Note 9

3. Estee Lauder's effective tax rate is 47.0% in fiscal year 2024. This is an increase over the past two years. Part of the reason for the increase is the nondeductible impairment charge.

   Which of the following is consistent with the impairment charge increasing the effective tax rate? Circle all that apply. (2 points)

   a. The impairment charge is not deducted from book income, but is deducted from taxable income
   b. The impairment charge is not deducted from taxable income, but is deducted from book income
   c. The impairment charge represents a temporary difference.
   d. The impairment charge represents a permanent difference.

   Not deducting something from income would increase the firm's tax expense and thus its effective tax rate. Permanent differences create a difference between the statutory and effective tax rates.

4. Consider Estee Lauder's 2024 fiscal year end deferred tax assets related to net operating loss carryforwards. Assuming that Estee Lauder capitalizes its tax-loss carryforwards using the 21% federal statutory tax rate, what is the total amount of future taxable income that Estee Lauder can offset using its net operating losses? (3 points)

   The DTA related to NOLs is $405 million. Assuming that tax loss carryforwards are capitalized at a 21% tax rate, the total tax loss carryforwards would be estimated at:

   Tax Loss Carryforwards x 21% = $405

   Tax Loss Carryforwards = $1,929

5. Estee Lauder's net Valuation Allowance increased between fiscal year 2023 and 2024. What effect would this net change have on fiscal year 2024 net income? (3 points)

   a. Increase $238 million
   b. Decrease $238 million
   c. Increase $38 million
   d. Decrease $38 million
   e. No effect

6. Assume that timing differences in depreciation and amortization are the only factor that creates a difference between the book basis and tax basis of fixed assets and intangible assets. What was the net difference between book and tax depreciation and amortization **in fiscal year 2024**? And which is higher? (as noted above, assume that deferred tax assets and liabilities are computed using a 21% tax rate) (5 points)

   Amount of the difference _____**$1,667**_____

   Circle one.
   a) Book depreciation and amortization is higher
   b) Tax deductible depreciation and amortization is higher

   The Deferred Tax Liability for Fixed Assets and Intangible Assets increased by $350 = $614 - $264

   DTL = accumulated temporary difference x t
   ΔDTL = Δ accumulated temporary difference x t
   = temporary difference x t
   350 / .21 = temporary difference
   1,667 = temporary difference = difference between book and tax depreciation

From our discussion in class, you could reason that tax deductible depreciation is greater than book depreciation. Otherwise, you can realize that, if the DTL increases it must be the case that:
Tax Payable < Tax Expense
Taxable Income < Book Income
Tax deductible depreciation > Book Depreciation Expense

7. Why would Compensation-related expenses create a Deferred Tax Asset for Estee Lauder? Circle one. (3 points)

   a. **Compensation-related expenses are expensed over the vesting period for book purposes, but are expensed at the time of exercise for tax purposes**
   b. Compensation-related expenses are expensed at the time of exercise for book purposes, but are expensed over the vesting period for tax purposes
   c. Compensation-related expenses are not deductible for tax purposes
   d. Compensation-related expenses are capitalized for book purposes

A DTA indicates that Taxes Payable > Tax Expense
Taxable Income > Book Income
Tax deductible Compensation < Book Compensation

This would be true if Compensation is delayed for tax purposes but expensed earlier for book purposes.

VI. Stockholders' Equity (26 points)

Refer to Estee Lauder Companies, Inc.'s 2024 financial statement excerpts to answer the following questions. **Note that the fiscal year 2024 is the year ended June 30, 2024.**

1. How much stock compensation expense did Estee Lauder recognize on the fiscal year 2024 income statement? (3 points)

   $325, see the SCF or Note 19

2. How much total compensation expense will Estee Lauder have to recognize over the vesting period for the stock options **granted in fiscal year 2024**? (3 points)

   See Note 19 Stock Option table for options granted x fair value at date of grant

   1,780.6 x $52.83 = $94,069

   This is the amount that will be recognized over the vesting period.

3. Which of the following is true with respect to accounting for Stock Options? Circle all that apply. (4 points)

   a. Compensation expense is recognized over the vesting period.
   b. Total compensation expense to be recognized is based on the fair value of the options on the grant date.
   c. Total compensation expense to be recognized is based on the intrinsic value of the options on the grant date.
   d. The expense that is recognized is adjusted over time when the company's stock price changes.
   e. The compensation expense recognized does not affect financing cash flows.
   f. No compensation expense is recognized when the options are exercised.
   g. Financing cash flows increase when the options are exercised.

4. How much did Estee Lauder **declare** in dividends in fiscal year 2024? (3 points)

   $954 million, see the Statement of Stockholders' Equity for the change in Retained Earnings. This is not the same as the amount of dividends paid ($947).

5. Why is Estee Lauder's diluted EPS less than Basic EPS for the fiscal year 2024? Circle all that apply. (3 points)

   a. **In-the-money Stock Options**
   b. **PSUs**
   c. **RSUs**
   d. Lower income
   e. Convertible Debt
   f. Convertible Preferred Stock

   See Note 21

6. If Estee Lauder declares a 10% stock dividend on June 30, how are the following affected? Assume that Estee Lauder's CEO earns a bonus by achieving a target ROE (Return on Equity), where ROE is measured as Net Income divided by Average Total Book Value of Equity. (4 points)

CHOOSE ONE

	Higher	Lower	No Change
Likelihood of Bonus	_____	_____	___X___
Retained Earnings	_____	___X___	_____
Stock Compensation Expense	_____	_____	___X___

A stock dividend is typically recorded as:

Dr. Retained Earnings
Cr. Common Stock
Cr. APIC

That is, just substituting Credits to Stock accounts for the Credit to cash in a regular cash dividend. Thus, total equity does not change and no income account is affected, but retained earnings is reduced.

7. How many of Estee Lauder's common shares are <u>outstanding</u> as of fiscal year end 2024? (3 points)

   From balance sheet: 596,560,598 (issued) – 237,871,995 (treasury) = 358,688,603 shares outstanding

8. What journal entry did Estee Lauder make to record the repurchase of its common stock during fiscal year 2024? (3 points)

Dr. Treasury Stock	**33**	
Cr. Cash		**33**

From the Statement of Stockholders' Equity and the Statement of Cash Flows: $33million

IV. Cost Concepts (30 points)

1. UM Company currently buys 15,000 units of a part used to manufacture its product at $39 per unit. The supplier recently informed UM Company that a 20 percent price increase will take effect next year. UM has some existing excess capacity and could produce the units themselves for the following per-unit costs (based on 15,000 units):

Direct materials	$18
Direct labor	$12
Variable manufacturing overhead	$10
Fixed costs	$8

If UM purchases the units from the supplier, UM can rent out the additional capacity in their plant and receive $22,500 per year.

Should UM Company buy the part externally or make it internally? Show your work for <u>any</u> credit. (5 points)

<u>Buy</u>:
15,000 x $39 x 1.2 = 15,000 x $46.8 = $702,000

<u>Make</u>:
15,000 x ($18 + $12 + 10) = 15,000 x $40 = $600,000

Opportunity cost of Making	**<u>$ 22,500</u>**
Total Cost of Making	**$622,500**

<u>Make</u> the units and save $79,500.

Fixed costs are not relevant because they are <u>fixed</u>. They do not change with the number of units or the make v. buy decision.

2. The Ibis Food Market is a merchandiser of organic food items. The company is considering the possibility of selling pomegranates for $0.30 each. Pomegranates can be acquired in unlimited quantities for $0.22 each. There are no additional variable costs associated with acquiring and selling pomegranates since labor is on a salaried basis. However, to acquire pomegranates at this price, Ibis Food Market must pay $1,000 per year for membership in an International co-op.

   a) How many pomegranates would Ibis Food Market need to sell annually to justify joining the co-op (break-even)? (3 points)

$0.3Q = $0.22Q - $1,000 = 0
$0.08Q = $1,000
Q = 12,500

   b) How many pomegranates would the company need to sell to earn a profit of $1,000? (2 points)

$0.3Q = $0.22Q = $1,000 = $1,000
$0.08Q = $2,000
Q = 25,000

3. Douglass Company produces only two products: a major computer part and a major cell phone part. The company expects to produce 4,000 computer parts and 2,000 cell phone parts in 2024. Each computer part requires 2 direct labor hours to produce, and each cell phone part requires ½ direct labor hour to produce.

   The direct material and direct labor costs included in the two products are as follows:

Cost	**Computer Part**	**Cell Phone Part**
Direct Material (per unit)	$30	$17
Direct Labor (per unit)	$16	$4

   Budgeted (Estimated) Total Factory Overhead Data for 2024: $270,000

   Required:

   A. Calculate the **total** cost per unit of each product assuming that the firm uses direct labor as the cost driver to allocate all factory overhead. (6 points)

   Computer Part **Total** Cost per unit: _____**$106**_____

   Cell Phone Part **Total** Cost per unit: _____**$36**_____

   Note that Computer part uses 2 DLH per unit and Cell phone part uses ½ DLH per unit. The firm makes 4,000 Computer parts and 2,000 Cell phone part.

   Plant-wide rate = $270,000 / (8,000 hours + 1,000 hours) = $30 per DLH

   Computer Part = DM + DL + MOH = $30 + $16 + $60 = $106 per unit

   Cell Phone Part = DM + DL + MOH = $17 + $4 + $15 = $36 per unit

B. Now assume that an outside cost consultant has recommended that the company use a more sophisticated activity-based costing system to charge overhead to products based on the following cost drivers:

Activity	**Budgeted Overhead**	**Computer Parts**	**Cell Phone Parts**	**Total Volume**
Production Setups	$80,000	5 setups	15 setups	20 setups
Material Handling	$70,000	1,000 lbs.	4,000 lbs.	5,000 lbs.
Packaging and Shipping	$120,000	4,000 boxes	2,000 boxes	6,000 boxes
Total	$270,000			

Calculate the activity cost rates for each cost driver noted above. (3 points)

a. Setups

$80,000 / 20 set ups = $4,000 per setup

b. Material Handling

$70,000 / 5,000 pounds = $14 per pounds

c. Packaging and Shipping

$120,000 / 6,000 boxes = $20 per box

C. Calculate the **total** cost per unit of each product using the activity-based costing system. (8 points)

Computer Part **Total** Cost per unit: **$30 + $16 + $28.5 = $74.5**

Cell Phone Part **Total** Cost per unit: **$17 + $4 + $78 = $99**

MOH per unit for Computer Part:

5 setups x $4,000	= $20,000
1,000 lbs. x $14	= $14,000
4,000 boxes x $20	= $80,000
Total	$114,000
	÷ 4,000
Total per unit	$28.5

MOH per unit for Cell Phone Part:

15 setups x $4,000	= $60,000
4,000 lbs. x $14	= $56,000
2,000 boxes x $20	= $40,000
Total	$156,000
	÷ 2,000
Total per unit	$78

D. Why did the cost estimates change the way they did, from the plant-wide method to the activity-based method? (3 points)

As the high-volume product, the Computer Part was allocated more overhead under the plantwide method, which used Direct Labor Hours as the cost driver, related to volume. However, moving to activities, it was clear that the Cell Phone Part used more of the costly activities, and thus it became clear that the Cell Phone Part is the more costly part to manufacture under ABC.

V. Transfer Pricing (16 points)

ABC Miners Operates two divisions, one in Japan and one in the United Kingdom. The Mining Division operates in Japan, which is rich in raw emerald. The Processing Division operates in the UK and processes the raw emerald into polished stone fit for human jewelry.

The cost details of these divisions are as follows (translated to British Pounds):

Division	Japan Mining Division	UK Processing Division
	Per carat of raw emerald	Per carat of polished emerald
Variable Cost	£50	£150
Fixed Cost	£100	£350

Several polishing companies in Japan buy raw emerald from other local Mining Companies at 9,000 Yen or £180 per carat (the current Foreign Exchange Rate is 50 yen = 1 Pound (£)).

Income Tax rates are 20% in Japan and 30% in the United Kingdom.

It takes 2 carats of raw emerald to yield 1 carat of polished stone. Polished emerald sells for £3,000 per carat.

1. Given the model we learned in class, what is the appropriate transfer price per raw emerald carat for the Mining Division to sell to the Processing Division? (2 points)

   a. Variable Cost
   b. Market Price, i.e., Variable Cost plus Opportunity Cost
   c. Full Cost plus a mark-up
   d. Dual Transfer Prices

   Because there exists a market price for Mining's goods, they would expect the market price as the transfer price so that they are not worse off for transferring internally.

2. Compute the transfer price of 1 carat of raw emerald transferred from the Mining Division to the Processing Division if the transfer price is set at: (4 points)

   a. 200% of Full cost

      £150 x 2 = £300 per carat

   b. Market Price

      £180 per carat

3. 1,000 carats of raw emerald are mined by the Japan Mining Division and then processed and sold by the UK Processing Division. Compute the *after-tax* income for each division using the **Market Price** as the transfer price. (7 points)

	Japan	United Kingdom
Revenue	**£180**	**£3,000**
VC	**- 50**	**- 150**
FC	**-100**	**- 350**
Transfer Price		**- 360** (£180 x 2 raw carats)
Pretax Income	**£30**	**£2,140**
Tax	**-6**	**- 642**
Income per carat	**£24**	**£1,498**
X carats	**x 1,000**	**x 500**
	£24,000	**£749,000**

4. Would setting the transfer price as the market price or 200% of full cost provide greater overall firm profit for ABC Miners? Circle one. (3 points)

   a. Market Price would provide a larger profit to the overall firm
   b. **200% of full cost would provide a larger profit to the overall firm**
   c. Choice of transfer price would not change overall firm profit.

   The firm pays less in tax when more profit goes to the lower tax rate country, Japan. The 200% of Full Cost transfer price is higher and thus would make the Japanese subsidiary more profitable.